Math 561 Assignment 2*

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Due date: 25 Jan 2026.

1.

(a) Verify the two polynomial equations in Example 1.1.2 of the textbook (Three-step Markov chain) which we discussed in class. That is, fill in all the missing details to verify the equations are a correct characterization of the Markov chain model.

(b) Does the point $\hat{p} = \left(\frac{20}{78}, \frac{4}{78}, \frac{18}{91}, \frac{3}{91}, \frac{10}{78}, \frac{2}{78}, \frac{24}{91}, \frac{4}{91}\right)$ lie in the 3-step Markov chain model? (This probability distribution shows up on page 7 of the textbook.)

2. Let $M \in \mathbb{R}^{m \times n}$ be an $m \times n$ matrix with entries in $\mathbb{R}$.

Prove that the following statements are equivalent:

(a) The rank of M is at most one.
(b) There exist vectors $u \in \mathbb{R}^M$ and $v \in \mathbb{R}^n$ such that $M = uv^T$, where "T" denotes the transpose.
(c) All 2×2-minors of M are zero.

Advanced: Work on the more general formulation. What analogous statement can you make for 3- dimensional tensors? what is a 'rank' of a tensor?

3. How does the mathematics of problem 2 relate to the model of independence of two discrete random variables?

(Advanced: Same question for three discrete random variables.)

4. Which points in $\mathbb{R}^2$ satisfy:

(a) theequation $p_1^2 + p_2^2 - 1 = 0$?
(b) the equation $p_1 - p_2^2 = 0$?
(c) toh equations (a) and (b)?

These solution sets of systems of polynomial equations are **algebraic varieties**. If S is a set of polynomials, then then the variety defined by S is denoted $V(S)$. You will see this concept appear early on in our textbook.

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